

SEPTEMBER 2024

Seeking Science VOL 38

SEEKING SCIENCE

MONTHLY NEWSLETTER

STEM Action Teen Institution



How is an AI model
compiled?

Kevin Zhang

AI Minor for Emory
Students

Brian Wang

Virtual Reality vs.
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Eddie Zhang

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Interdisciplinary AI Minor for Emory Students

Featured Article

Brian Wang

AI has improved at an alarming rate in numerous ways: text generation, idea generation, code editing, generation for study guides, and much more. Within the next decade, it is likely AI will replace many current jobs on the market. Many top universities are creating AI courses to combat this for the survival of their students in the future. One of the many methods is an AI minor open to all students of all majors at Emory University. This AI Minor course will highlight the definition of AI, how it is used, its goals, intentions, and ethical issues with using AI.

Those at Emory believe that AI is going to be crucial to all parts of society, and therefore recommend that students learn about its various components: how bias enters algorithms, how AI generates text, how AI takes past data and combines it with the current context to produce a result, and much more.

Although not many universities are geared with majors or minors in AI yet, many top schools are already taking their rankings in artificial intelligence. For example, Carnegie Mellon University is #1 in Artificial Intelligence as of September 2024, with Stanford and MIT taking second



and third place. A student from Princeton developed the technology to predict whether a piece of text is AI-generated or hand-written.

AI will impact not just the computer-related fields, but also those that deal with literature or music. At its core level, AI is made from computers, so it can do computer functions such as basic operations and even code. Currently, it cannot produce complex code at a proficient level, but it may very well do so in the coming years. On other levels other than computers, many AI technologies have already branched out to create deep fakes of famous artists voicing famous songs. Some produce music with AI-generated lyrics (which are sung by AI too). The literature business also has instances where people utilize AI to write articles for money - or stories at that.

With the introduction of AI evolving in the education system out of the way, how does AI work? From the level of AI-generated text detection, the main two metrics that are used are perplexity and burstiness. Perplexity measures how well an AI language model can predict the next word in a sequence based on current data in its “database”. Burstiness refers to how unique the language is used throughout the text. A language model typically uses the same level of burstiness in its language throughout a passage. Humans vary in vocabulary while writing articles and thereby create more burstiness.

With coding problems and music, AI typically uses past data to train itself to generate new data. For example, many AIs answer questions about Leetcode (a popular complex programming website) by training on solutions within the website and even some Stackoverflow discussion forums for the programming question. Often, the AI answers similarly to those on the discussion forum.

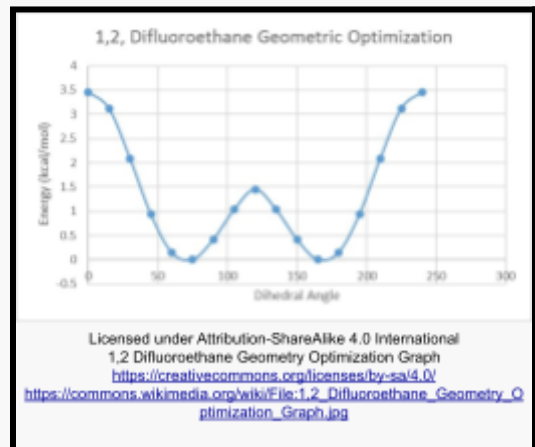
Whatever the case, AI has already taken a hold on our community. It has already branched out to industries such as math, programming, reading, music, counseling, finance, etc. As citizens of the future, we must learn about how such an important tool is evolving and utilize it ourselves for the best optimal output.

How is an AI model compiled?

Kevin Zhang

Specializing an AI model entails taking a conceptual description of the model written in a language such as TensorFlow, PyTorch, or Keras and translating this down to the required hardware shape. The abstract model architecture is transformed from compilation into the target system it will run on, be it CPUs, GPUs, TPUs, or even purpose-built AI processors. This process involves several steps: being focused on enhancing graph structure, reducing arithmetic precision (for example, from floating point to integer arithmetic), and emitting target ISA-specific code.

The first step is called graph optimization during which the computation graph of the model is optimized for pruning, fusing, and re-ordering. For instance, cutting out duplicated tasks could be performed, and synchronous layers can be combined. This draws down the model's size and the computational requirements, which is essential in situations where the model is meant to work in real-time or the hardware where it will be deployed has limited processing power such as in the case of mobile phones or embedded systems.



Precision reduction is another strategy implemented mostly in edge AI strategies, followed next. Rather than performing the calculations using 32-bit floats which is normal during the training of the model, the model can be quantized to work with 16-bit or 8-bit integers. This cost trade-off which is accuracy against time and computational resources ushers in efficiency in model computations which always come with reasonable accuracy loss.

The model is thus translated into a low-level intermediate code (Low-Level IR) or directly into hardware-dependent code. For instance, in TensorFlow, the model may be transformed to TensorFlow Lite when it runs on a mobile application. In other cases, there can be predefined custom compilers such as TVM, XLA, or OpenVINO that can create unique low-level binary code that can be run on the target system. Optimizers may be CPU/GPU/accelerator specific, and it is possible to utilize NVIDIA's TensorRT as part of the compilation pipeline.

Last, there is the compiled model that is bundled with runtime libraries for managing the device's operations including scheduling memories and paralleling the execution. Inference inference-ready environment of the target platform is now created based on the compiled model. Through these optimizations, it is possible to get close to the peak performance for a given task like detecting objects, natural language processing, or real-time data analysis and this kind of AI model can be implemented in a variety of devices and environments.



Virtual Reality vs. Augmented Reality

Eddie Zhang

Virtual Reality (VR) and Augmented Reality (AR) are two technologies that blend the digital and physical worlds in different ways, offering unique experiences and applications.

VR immerses users in a fully digital environment, replacing the real world entirely. Using headsets like the Oculus Rift or HTC Vive, VR transports users into a simulated reality where they can interact with digital objects as though they were real. This makes VR particularly valuable in gaming, virtual tours, and training simulations. For example, medical students can practice surgeries in a risk-free virtual environment, while gamers can experience a fully immersive gaming world. However, VR's immersive nature can also be isolating, disconnecting users from the physical world and sometimes causing discomfort like motion sickness.



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AR, on the other hand, overlays digital information in the real world, enhancing the user's perception of their environment without replacing it. Devices like smartphones, tablets, or AR headsets like Microsoft's HoloLens project digital content onto the physical world. AR is practical for applications like retail, where customers can visualize products in their homes before purchase, or in navigation, where users can see directions superimposed onto their real-world surroundings. Unlike VR, AR is more accessible, as it can be used on common devices, but it depends on the real-world environment, making it less effective in poor lighting or cluttered spaces.

The key difference between VR and AR lies in immersion. VR offers a completely immersive experience, ideal for scenarios that require full focus on a simulated environment. AR enhances reality, making it more suitable for practical, real-world applications. As these technologies evolve, they will continue to complement each other, shaping how we interact with both digital and physical worlds.



The Next Frontier: Exploring and Colonizing Space for Humanity's Future

Brandon Wang

The Moon, Mars, and beyond are our first objectives on the way to where humanity will ultimately live in space. The reasons for this drive are numerous, ranging from the hunt for fresh commodities with which to ensure mankind's modern welfare and even potentially establish extra-terrestrial colonies. Hence, humanity ensures its survival by diversifying its presence beyond Earth.

With other governments and companies also exploring space, progress is rapidly accelerating. As mentioned, government space agencies such as NASA and ESA have been the dominant players conducting missions to better understand planets, asteroids, and more. Over the past few years, however, that has changed with private companies such as SpaceX and Blue Origin entering the scene with lofty goals around human trips to Mars or commercial space travel. Public-private partnerships are sparking innovation and cutting the cost of space missions.

The Moon and then Mars are likely the two most feasible colonization candidates in our near-term future. A lunar base is regarded as the vital precursor to further exploration, and potential economic usage of the Moon's resources. Mars is widely considered to be the next Earth for human settlement by some researchers and acknowledged as having biotic hopes due to its similitude with our planet in terms of conditions. Scientists have already planned missions to Mars for the next decade.

Space colonization is fraught with technological and logistical challenges, some of which seem impossible to solve. There are still a myriad of problems that need solving before it will be possible to live off the world: from protecting humans against space

radiation to developing self-sustaining life support systems. There are also major ethical considerations such as planetary protection and space governance. When colonies are established how resources will be shared, and who sets the rules of the road for settlements in space will likely figure large.

Thus, space colonization is not an odd or distant ideal for all human beings. By confronting the challenges we make, innovations are possible and foster cooperation both between nations and within our society. We still have obstacles to overcome, but the dream of expanding human civilization from Earth and ensuring humanity among the stars is quickly rising as a priority for scientists, engineers, and leaders around our world.



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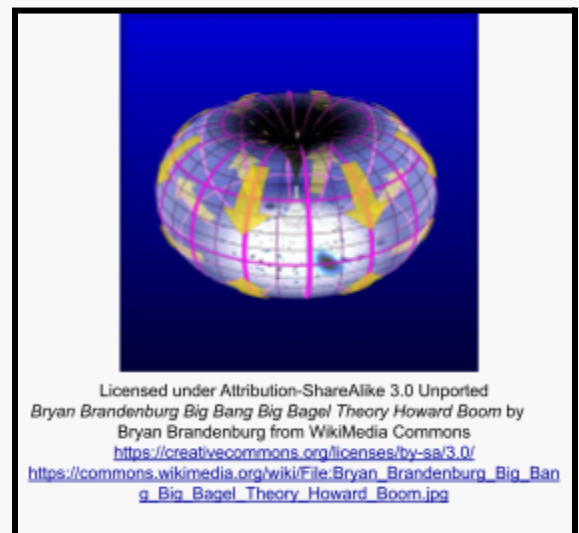
Is the Universe a Doughnut?

Yidian Wang

Have you ever wondered what the universe's shape is, if it even has one? The hypothesis is that the universe is shaped like a doughnut; an Euclidean 3-torus. is a fascinating theory in cosmology. Many people may think the universe is infinite and ever-expanding with no end, and though this may be true, the universe being a doughnut would make more sense.

Doughnuts have a complex shape. You can travel across the entire object and end up back where you started, much like a sphere. Earth is in the shape of a sphere, so if you walk in a straight line from one point you will eventually travel the entire globe and end up in the same spot. The loop can be shrunk to a single point because it can be traveled in a closed loop. If the universe has a doughnut shape, you could travel the whole cosmos and end up in the same spot. But unlike the Earth, you cannot squish it to a single point if you travel a loop around the hole because the hole gets in the way. This makes the doughnut a complex shape, or a “nontrivial topology.”

Scientists are still looking for signs of the universe looping back on itself. They've looked at the cosmic microwave background, essentially searching for identical shapes of light that appear in different places in the sky, and for similarities between different areas. Although there's still no proof that the universe is in the shape of a doughnut, the boundless complexity of the universe and our minimal knowledge of it allow for more improvements and discoveries to emerge in the future.



The Evolution of Lighting

Ben Liang

The sun is the most important thing on earth. It is the driver of a lot of the earth's systems including light. Without light, humans wouldn't be able to see and that's why humans have been trying to create a light source. Light sources for humans need to be at least 10 to 20 lumens for humans to see. Light back then was different than now.

Natural light is the most abundant light on the earth. It is around 6000 to 10000 lumens on average. Ancient and modern people have made houses that can light up houses when the sun rises.

Torches are the next invention that humans made for lighting. They are around 16 lumens and can burn for 20 minutes to 1 hour. The candle is used for a very long time. It was made 3000 to 5000 years ago. It was made with animal fat and grease. It had an average of 13 lumens and burned for 10 to 13 hours per inch of diameter. Gas lamps were next that were made. It was made in the 1790s. It only produced 2 lumens per watt.

Next were electric lamps. It was made by Thomas Edison and had around 700 lumens. After that, a few electricity shortages caused the first fluorescent bulbs to be made. It was made by Heinrich Giessler and Julius plücker. They had 50 to 100 lumens per watt. Leads came after that. The led was invented by Nick Holonyack and it made 75 to 110 lumens per watt.

Light is a very important resource. It was made by biotic resources at first and it switched to only using some. In conclusion, light is important now and back then, but we make it more efficient now.

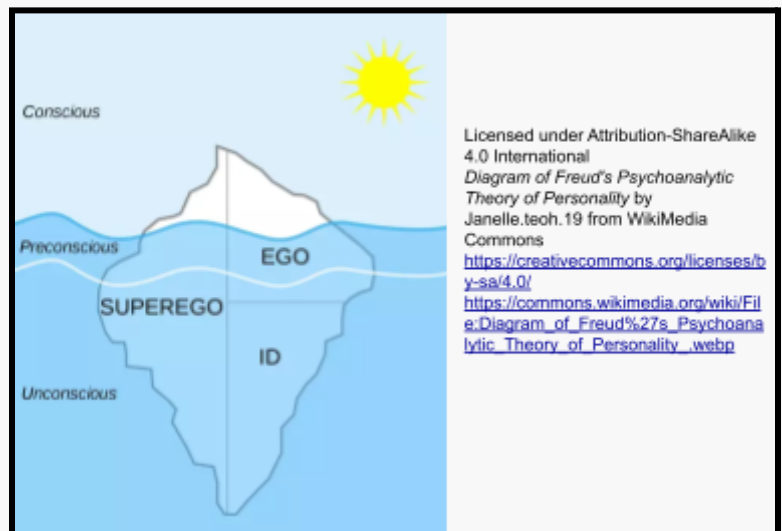


Freud's Dream Theory

Richard Wang

Dreams, a fascinating phenomenon all happening in our sleep, are very significant for understanding the unconscious. Sigmund Freud, the founder of psychoanalysis, set out to propose a theory about dreams. It also started on the night of July 24, 1895, in a dream, when his patient wasn't responding well to a treatment as he had anticipated. Plagued by guilt, he blamed himself. However, it turns out that her poor response was because of a dirty syringe administered by another doctor and this freed Freud's guilt. Subsequently, he proposed that his dream was a fulfillment of the wish that the patient's sickness was not his fault. He went on to explain his story in the book called *The Interpretation of Dreams*.

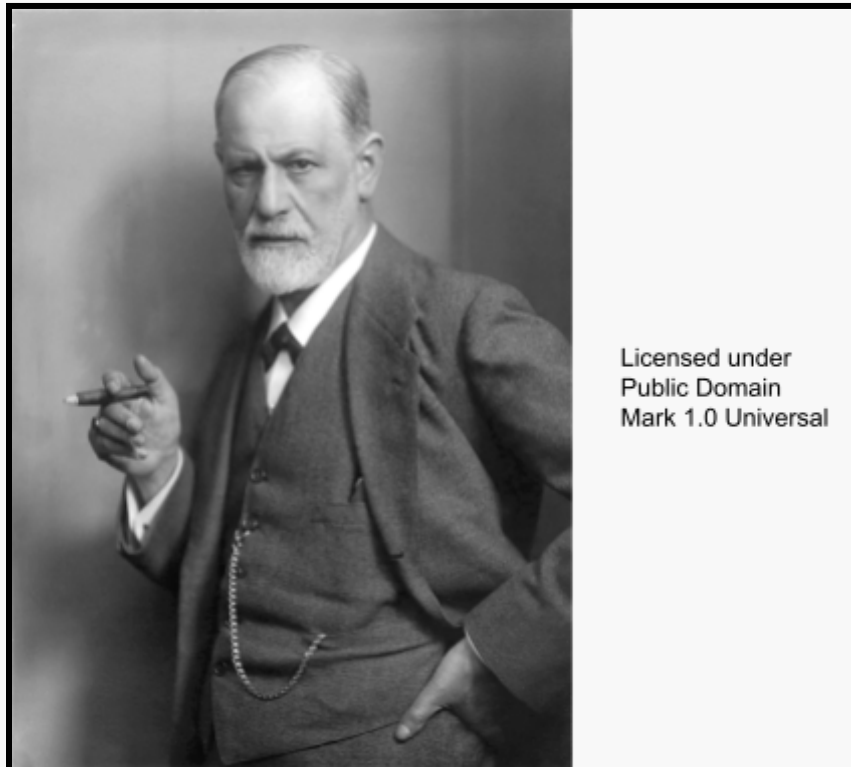
According to Freud, dreams arise from two different mental processes. In the first, the unconscious mind constructs a wish to be expressed. Then, the second process censors the expression of the wish to prevent them from being fully conscious because they might conflict with moral values and our consciousness. This distortion makes the dream feel like a blur rather than a straightforward representation. Freud also advanced the idea of manifest content and latent content.



Manifest content is the actual images that play out in the dream while the latent content

is the hidden meaning of the dream and consists of the dreamer's hidden thoughts or desires.

Freud claimed that dreams come from an experience from the previous day. He described four categories of content that people dream about. The first is the direct expression of mentally significant experiences. The second is recent and important experiences being combined into a single storyline. The third is a seemingly trivial dream that symbolizes a significant event. The last is a powerful memory that is symbolized by a seemingly insignificant mental event. People who experience external stimuli in their dream, such as an alarm clock, experience this because the mind is shut off from the real world and so our mind ends up incorporating that stimuli into the dream.



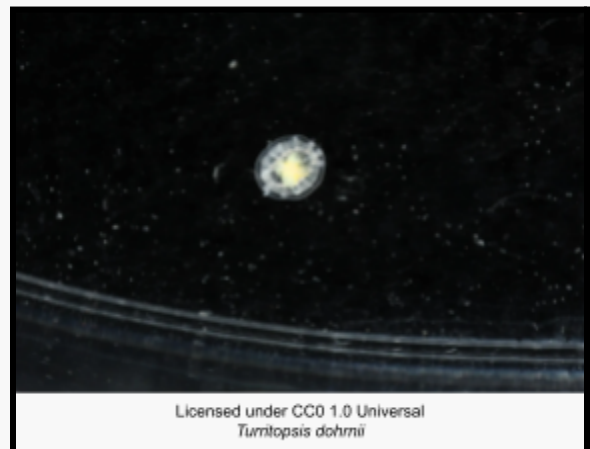
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The Immortal Jellyfish

Kenny Wu

Immortality is controversial, yet irresistibly tempting to everyone. Modern scientists have come far in research for immortality, however, still unable to perfect technologies such as CRISPR to provide a safe method of genetic editing. In their quest of searching for immortality, scientists came across an astonishing species of Jellyfish, namely *Turritopsis dohrnii*.

To begin, the immortal jellyfish, otherwise known as *Turritopsis dohrnii*, is a remarkable organism that captivated scientists for its trait of being biologically immortal. While most other living beings bear the fate of death, the immortal jellyfish can refurbish itself of age, returning to its initial form of juvenile polyp. The process is known as transdifferentiation and allows the biological immortality of *Turritopsis dohrnii*.



Furthermore, the process of transdifferentiation can be triggered by factors outside of old age. *Turritopsis dohrnii* will reset their adult body back to a young state under stress and physical damage. The jellyfish is capable of fully restarting its life cycle and regenerating its body, reflecting their superb defensive capabilities. *Turritopsis dohrnii*'s ability to revert its body state and defy the aging process is completely exclusive and unheard of.

All in all, the concept of immortality is soon to be grasped with the discovery of these immortal jellyfish. Scientists are spending a great effort to unravel the genetic and molecular mechanisms behind the jellyfish's transdifferentiation process.

The Importance of Sleep

Mary Liang

Sleep has always been an essential part of human survival, so if one doesn't get enough sleep, unfavorable outcomes tag along. For a healthy adult, it takes 15-20 minutes to enter the first of the three stages of sleep. Falling asleep too fast can indicate signs of a wearied body since the body naturally gives one a stronger drive for sleep. Sleep debt is also something to consider. If there are several days that one sleeps less than required, the body would sleep more than usual, this can last for multiple nights.

When one enters the sleep cycle, it can be presented that there are 4 different

sleep stages, preferably 4

to 6 cycles for quality

sleep. The first one is light

sleep, at this time, the

muscles relax and the

breathing slows down.

This stage goes on for 1 to

5 minutes. In the second

stage, the body

temperature goes down

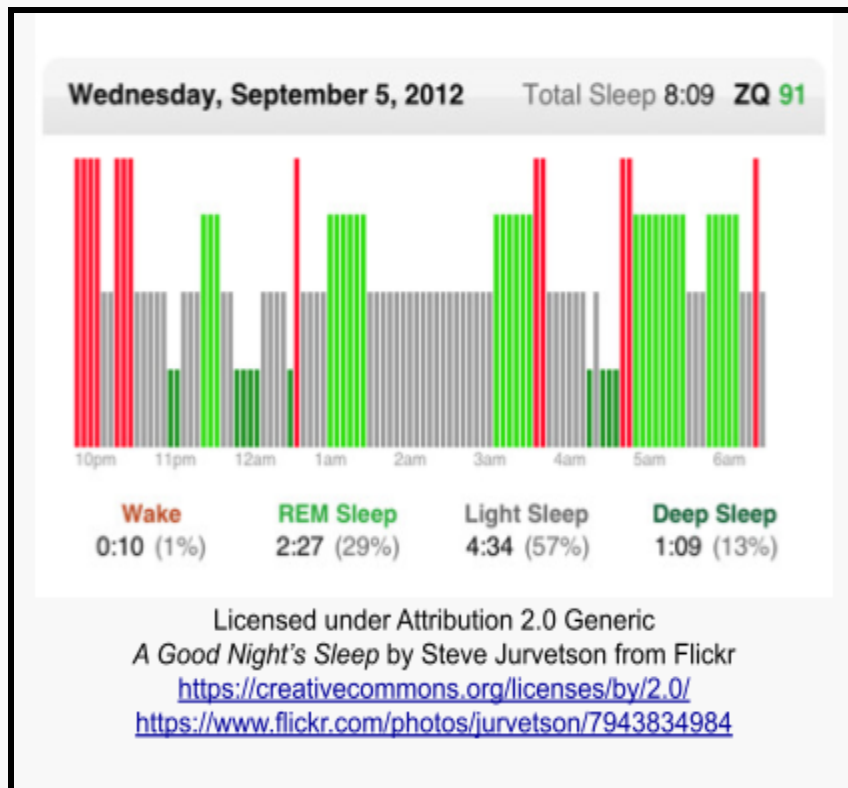
and the heart rate slows

down and lasts around 25

minutes. The third stage is

when the body builds

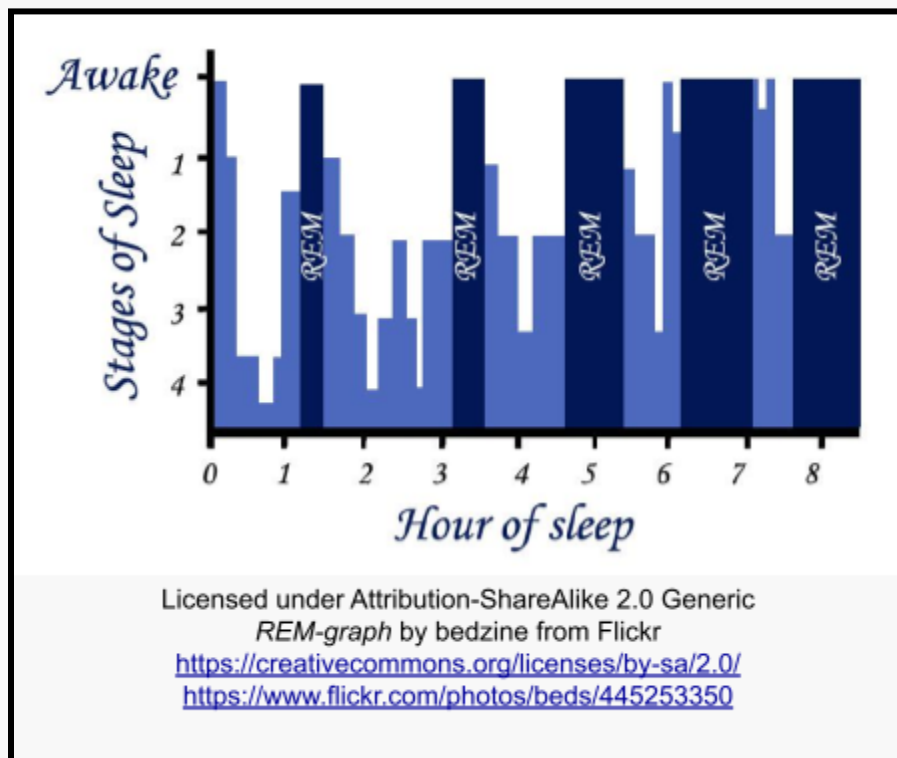
bones and muscles.



Anyone awoken at this stage experiences slightly decreased mental performance. The last stage is called REM sleep. This stage is also known to be the deepest stage of sleep,

making it harder to wake one up, and also the stage where the brain is highly active. Irregular muscle movements, an increased amount of oxygen intake, and vivid dreams can be found in REM sleep.

As teenagers grow into adults, their biological clock forces them to sleep about 2 hours after their original sleep schedule. They require between 8 to 10 hours of sleep every day. And with the fact that high school might start earlier than middle school, it is not impossible for them to not get enough sleep daily. This would affect their brain functionality and the increased chances for them to make irrational decisions and their future might revolve around bad habits that could interfere with their sleep, like being exposed to blue light before bed, the sleeping area being too hot or cold, etc. In conclusion, sleeping is as essential as eating and drinking water so it is important to get a good night's sleep.



Effect of Ocean Acidification

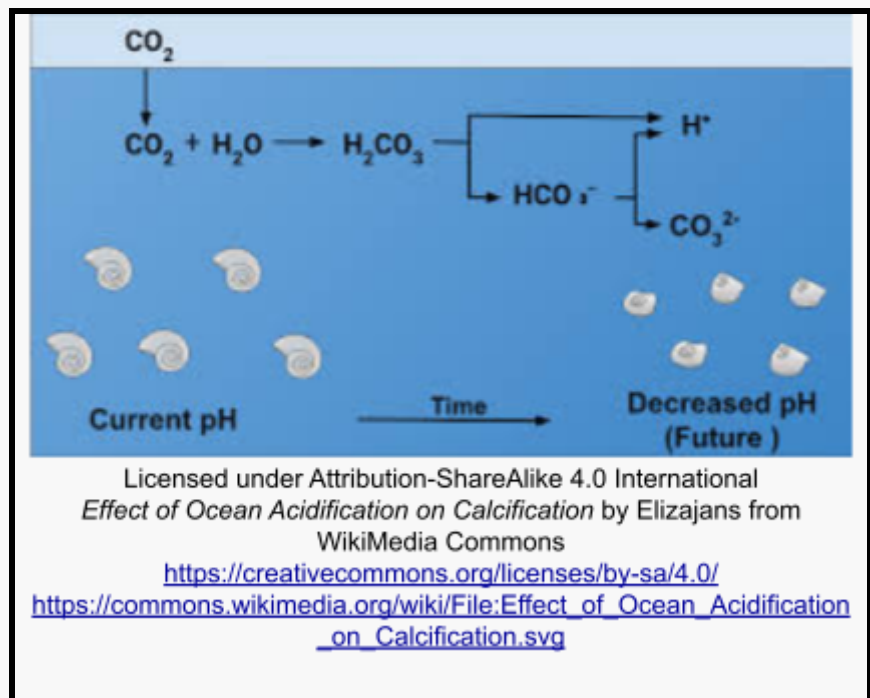
Owen Chen

From the start of the industrial revolution, the concentration of carbon dioxide had been rising due to the high quantities of fossil fuels burned daily. About 30 percent of the atmospheric carbon dioxide gets recycled into the ocean. Though it sounds beneficial, the constant observation of carbon dioxide is causing the seawater to get more and more acidic, creating an environment that is becoming more unsuitable for marine animals to dwell in.

To begin, the high density of carbon dioxide in the atmosphere didn't just appear out of nowhere like magic. According to the law of conservation, matter cannot be created nor destroyed, meaning the matter is recycled through other forms. One way to view the rising of carbon dioxide is from anthropogenic events.

Ranging from industrialization to deforestation, humans are constantly transferring carbon dioxide at a rate where Earth is unable to

cycle. This overloaded concentration has provided many negative consequences and is one of the main causes of loss in biodiversity.



As said before the high concentration of carbon dioxide had resulted in a decrease in pH level throughout the ocean. This is through the process of ocean acidification, where the carbon dioxide begins to break off hydrogen ions when contacting the seawater, eventually ending up as carbonic acid while also reducing the amount of carbonate ions. Carbonate ions are the building blocks for forming skeleton-like structures for corals and seashells. With the seawater getting more acidic and an increase in carbonic acid, building and maintaining the calcium carbonate bonds has become a challenge, which also affects smaller marine lives since they are dependent on these calcifying organisms for shelter and protection.

Ocean acidification also harms non-shell-categorized marine life. The increased acidity of seawater had dulled the sense of many prey to detect nearby predators. As a result, this can lead to a long-term effect of increased growth rate for predators, while the decline of smaller fish. Eventually will put the entire marine food chain at risk, leading to an Earth without any marine life.

Ocean acidification has been affecting Earth for centuries. With many species perished and the continuous rise of humans, it had become more difficult to maintain the ocean's homeostasis. Therefore, humans should be aware and more conservative as many economies are dependent on the ocean to thrive.



CRISPR: The Future of Gene Editing

Natalie Dai

CRISPR tech makes gene editing as simple as cutting and pasting text on a computer, which is interesting. This groundbreaking tool allows scientists to make, precise, changes to DNA the, life's blueprint.

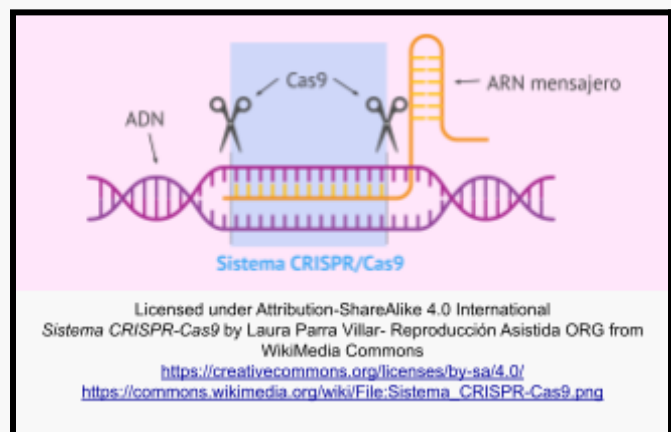
Clustered Regularly Interspaced Short Palindromic Repeats, abbreviated as CRISPR, represent a groundbreaking gene-editing technology that can profoundly change medicine, agriculture, and biology. Initially discovered in bacteria, CRISPR acts as a natural defense these microorganisms use against viruses. Realizing the potential of this system, scientists leveraged it to target and edit specific genes in plants and living beings.

Its functionality can be likened to a pair of molecular scissors. A protein known as Cas9 is guided by RNA slices, and DNA at a designated spot. Following this cut researchers can disable a harmful gene, or fix a genetic mutation, — or they might insert new DNA into the genome.

CRISPR has already shown potential in treating genetic diseases; a case in point involves correcting the mutation that causes sickle cell anemia using CRISPR in human cells.

A broad range of diseases like cystic fibrosis and certain cancers might be fixed with this technology.

The power to change genes raises serious ethical issues: should we change the genes of human embryos which could change future generations? CRISPR might be used to create "designer babies" with greater intelligence or improved looks.



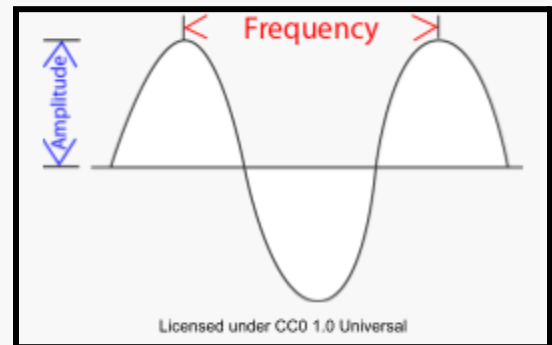
Sound Waves

Riley Lee

Sound waves are mechanical waves that travel with vibrations through air, water, and solids. Mechanical waves are waves that need a medium such as air or water to move. Sound waves are also longitudinal waves which means that the medium particles move back and forth in the same direction the wave travels. When a sound wave moves through a medium, compressions and rarefactions are formed. Compressions are areas where the particles are close together while rarefactions are areas where the particles spread apart.

Sound wave movements and the noise they create are based on key factors such as the frequency, the wavelength, the amplitude, and the speed. The frequency in sound waves is the amount of compressions and rarefactions happening in one second. Frequency is measured in a measurement called Hertz (Hz) and it determines the pitch of the sound wave. High-frequency waves create a high-pitched sound such as a whistle, while low-frequency waves create a low-pitched sound such as a drum. Wavelengths also play a part in frequency as lower and shorter wavelengths mean higher frequencies, but longer and bigger wavelengths mean lower frequencies.

While frequencies determine the pitch of the sound wave, the amplitude determines the loudness of the wave. The larger the amplitude, the louder the sound, but the lower the amplitude, the lower the sound. The speed of sound waves depends on the medium it travels through. Sound waves travel faster through solids than in liquids and faster in liquids than in gases.



The Future Of Photosynthesis

Brandon Pian

In the modern day where more and more things can be made artificially with the help of science and photosynthesis is one of them. Photosynthesis is the process plants use to convert sunlight into glucose for energy. Artificial photosynthesis was first proposed in 1912 by Italian chemist Giacomo Ciamician and was first tested in 1983. Artificial photosynthesis mimics natural photosynthesis in its process of creating energy.

Artificial photosynthesis uses solar energy to produce hydrogen from water. This hydrogen is then combined with CO₂ from factories to produce olefin. A more detailed explanation of this process is first splitting water into oxygen and hydrogen using sunlight and a photocatalyst. Then they use a separation membrane to split hydrogen and finally, they use a catalyst to create a reaction between hydrogen and CO₂ to produce olefin. Catalysts are crucial to accelerate the chemical reactions to make the processes faster. Catalysts are made up of different types of metals and sustainable materials like metal-organic frameworks.

Artificial photosynthesis is used to create chemical energy in the form of carbohydrates from sunlight like normal photosynthesis. The energy artificial photosynthesis produces is stored in chemical bonds making it be more easy to store and transport than solar-powered energy. Artificial photosynthesis can also remove excess carbon dioxide from the air, reducing the effects of global warming. Since artificial is still new it isn't fully complete, letting natural photosynthesis be still more efficient than artificial photosynthesis. A major challenge of artificial photosynthesis is scalability, with the cost As a result of these problems, artificial photosynthesis is still not widely used.



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Joint Center Artificial Photosynthesis (JCAP)

How do you optimize multiple AIs running at once?

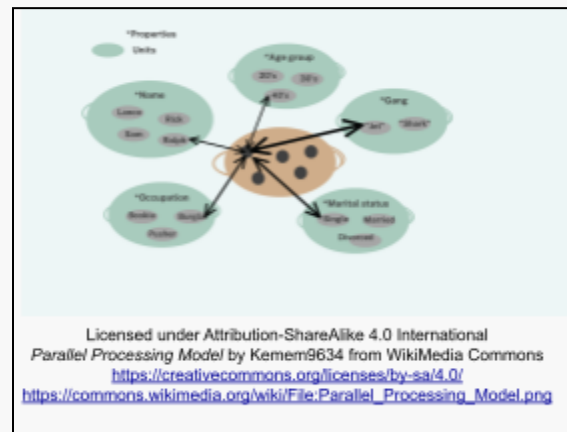
Audrey Don

As our societies improve every day, questions come with the improvement in our societies. For example, AIs, AIs are simulations of human intelligence processes by machines especially from computer systems. Learning, reasoning, problem-solving, perception, and language are all examples of cognitive abilities. People would ask how we can optimize multiple AIs running at once. Running multiple AI systems at once in a single operator, especially in a complex environment, requires many resources, especially management.

In this environment, it involves processing power, memory, etc. It would also require cloud computing and parallel processing techniques. Unlike us humans, AI systems aren't able to experience "burnout" in a psychological sense, but they can suffer from bugs in system performance from hardware failures, software bugs, or algorithmic inefficiencies. However, as AI systems become more autonomous, questions about

accountability, it's essential to establish clear guidelines for AI development, including ethical AI frameworks, rigorous testing standards, and policies for oversight and control.

The AI's risk of failure, where an AI's breakdown leads to a wider range of system collapse, can cause serious concern in our societies. In an environment where AI's used just about everywhere online, different AI working along with multiple operators at once could end up collapsing the system altogether.



How Spiders Avoid Get Stuck On Their Webs

Denise Lee

If spider webs are so sticky they can trap insects, how do spiders avoid getting trapped? Scientists are still not entirely certain how most spiders prevent getting tangled in their web, but they do know that spiders can spin different kinds of silk as not all of their silk is sticky, web-spinning spiders have an extra set of claws on their feet, they have tiny hairs on their feet, and their cautious movements. The spiders know which threads to avoid because only the sticky silk is dotted with glue.

The spokes of the web are not sticky, while the strands that circle the spokes are made from sticky silk.

The spiders know the difference between the two webs to avoid the sticky ones. All spiders have two claws on their feet; web-spinning ones have three. These claws are used to grasp threads and provide traction as the spider moves.



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Their moveable claws on their feet grip and release the web's threads if they step on the sticky web. Tarsi are the hairs on spiders that reduce the amount of surface area that

touches the silk. These hairs are also seborrheic and can also serve to provide a layer of anti-stick Teflon. The legs of spiders are also lubricated by oils responsible for enhancing the ability of the spiders not to stick to the viscose they produce to form their webs.

They also know how to maneuver and move in such a way that avoids the sticky parts of the web. A spider is sensitive to vibrations on the web or even the ground on which it is situated and uses such vibrations in hunting its prey and at the same time avoiding its webs. Altogether, these mechanisms allow spiders to move effortlessly across their intricate webs without getting tangled.



How are Oyster Pearls Made?

Arthur Liang

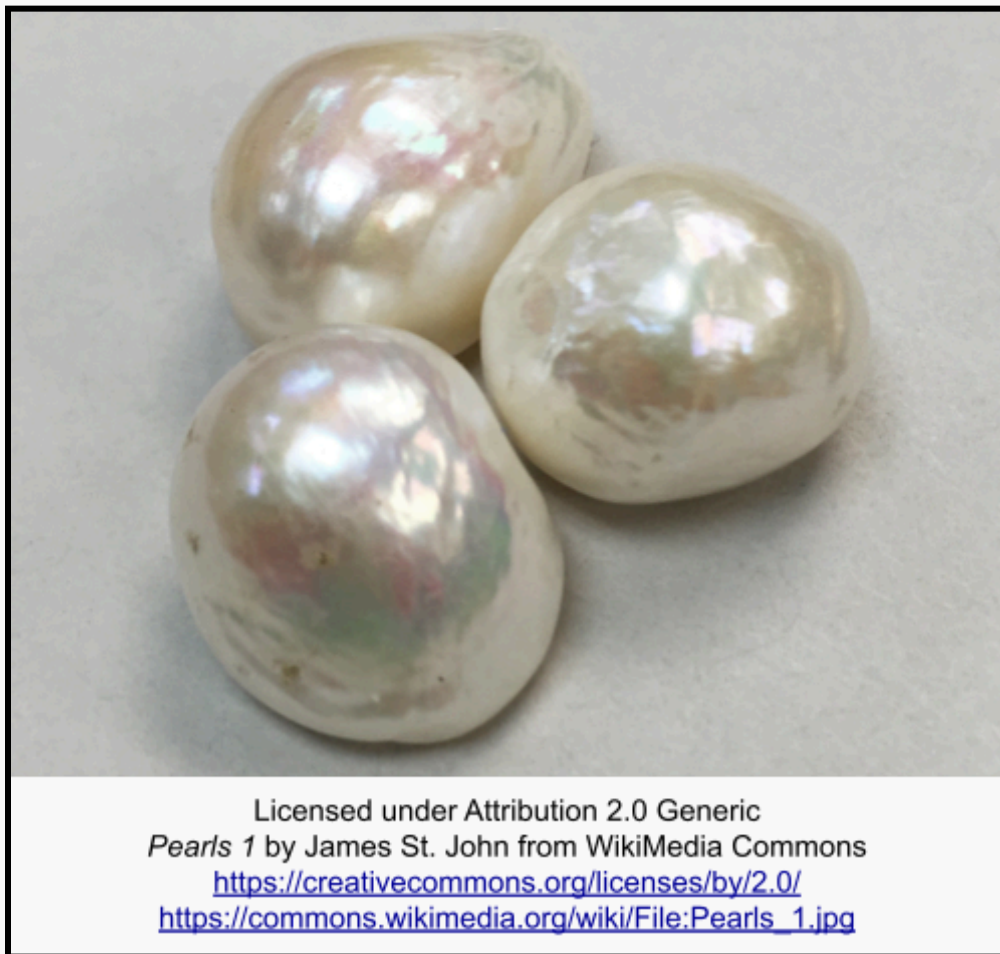
Although they appear unassuming, oysters and other mollusks have a unique ability seen nowhere else in the animal kingdom. Inside their shell, occasionally you may come across a shiny, lustrous pearl. These pearls vary in color, ranging from white to pink to silver and even to black. It seems almost impossible for an organic organism to form a mineral so perfect and attractive, so how do oysters do it?

The process begins when an irritant enters the oyster. This can be a grain of sand, a small parasite, or any other potentially harmful foreign object. The creation of the pearl acts as a defense mechanism. To protect itself from the irritant, the oyster begins secreting an iridescent substance called nacre. Nacre, also known as mother-of-pearl, is made up of a form of calcium carbonate known as aragonite, and conchiolin, a protein that acts as a binding agent. Over many years, the irritant is layered with nacre. The longer the process is allowed to continue, the bigger, smoother, and shinier the pearl becomes.



The variety of colors that pearls come in is determined based on the oysters themselves and their surrounding environment. Habitat conditions and minerals present in the water can affect the physical appearance of the pearl created. For example, the black-lipped oyster, found in the South Pacific, produces the famous Tahitian black pearls by using the same nacre which is used to line the inside of its shell.

Most pearls created today are cultured, meaning they are manually farmed by purposefully inserting a small bead into the oyster and waiting for the pearl to materialize. Natural pearls are extremely rare and highly prized, making them exceptionally valuable. Whether cultivated or natural, the formation of the pearl is a fascinating chemical process, a testament to the diversity and complexity of nature's creatures.



X86 vs ARM

Aidan Hong

Every computer has a processor to compile code, run programs, and more. However, these processors are all run on some kind of architecture. Two of the most common chip architectures out there are x86 and ARM. While both accomplish similar tasks, they're vastly different and natively incompatible.

x86 has been around for a much longer time compared to ARM. Many PCs and workstations use x86, and so do older Macs. x86 also supports legacy software and since it has been around for a long time, older operating systems use x86. It's also one of the many reasons why many consumers choose x86 PCs: its ability to run older software.

Compared with x86, ARM is much newer. Historically, ARM is typically used on mobile devices due to their low power consumption compared to x86 devices. However, just because they have a low power consumption doesn't mean they're not powerful! Apple's latest Macs have switched from x86 to ARM, and not only do their Macs now use less power, but they're also more powerful than their previous x86 Macs. Despite this, a huge downfall of ARM devices is their inability to run x86 programs. Apple has attempted to solve this with Rosetta 2, a program used to translate x86 programs to ARM, but it still isn't perfect, and not all programs can use Rosetta 2 to run on M1 Macs. For example, a VM designed to run on an x86 computer cannot run on an ARM Mac due to its vastly different architecture.

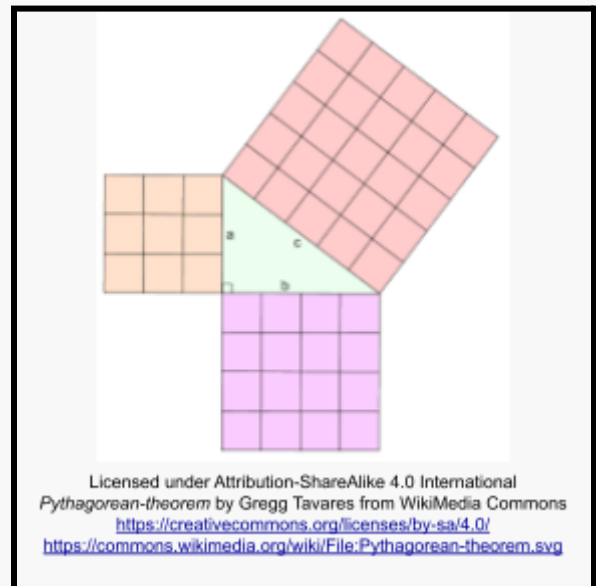


What is the Difference between Trigonometry and Pythagorean Theorem?

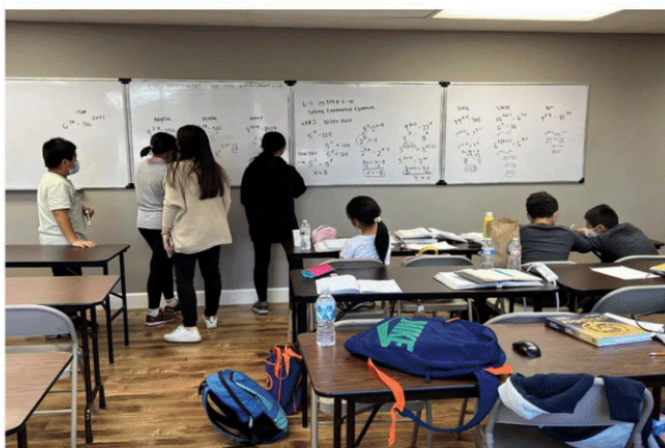
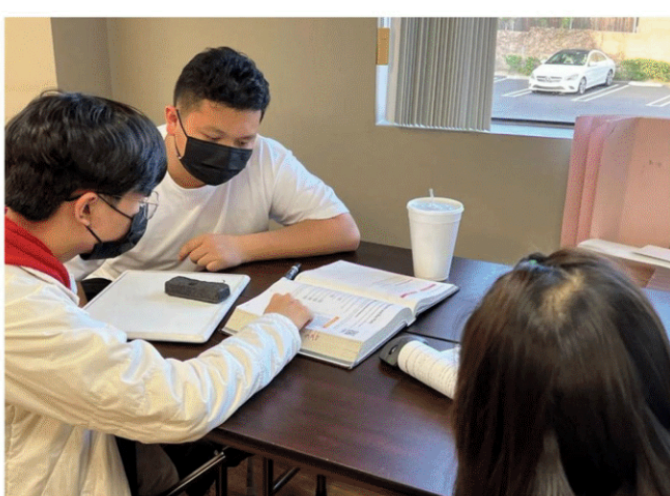
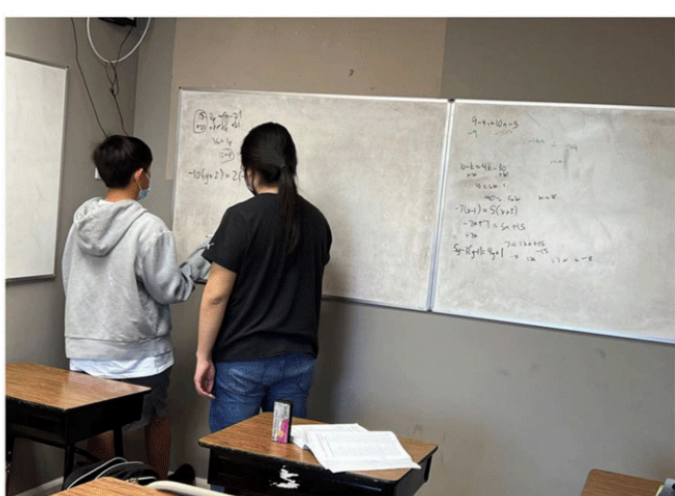
Emily Ma

The Trigonometry theorem calculates the lengths of triangle sides, and the Prothgram theorem studies the relationship between the sides. They both sound very similar, but they are entirely different. For example, you can quickly solve for the other side when solving for a triangle's length and knowing one angle and one side. However, in the Prothgram theorem, you must know at least two sides to solve the other. Also, it would help if you had to know which format to use when solving the problem math because you might end up solving for the wrong thing and having to start over.

In mathematics, the Pythagorean theorem or Pythagoras' there is a fundamental relation in Euclidean geometry between the three sides of a right triangle. It states that the area of the square whose side is the hypotenuse is equal to the sum of the areas of the square on the other two sides.



However, in trigonometry, the formulas are tan, sin, cos, and so on. There are many different ways to calculate angles and sides; it is a branch of mathematics concerned with relationships between angles and side lengths of triangles. In particular, the trigonometric functions relate the angles of a right triangle with the ratios of its side lengths.



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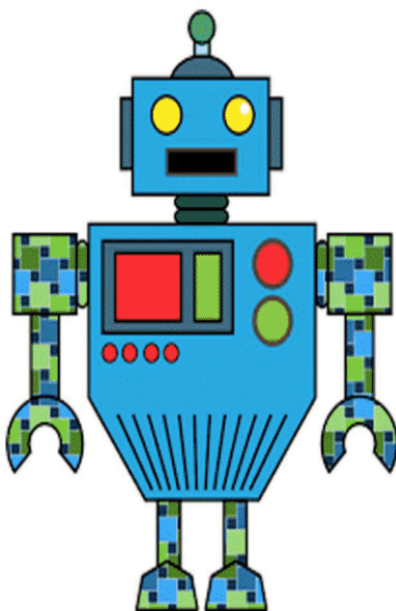
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